Version 2026-04-08 23:05, Eric Pohl

9. Course work

Day 1 ----------

- **Welcome (8:00-8:15)**
  - Aims
  - Expected outcomes
  - Wishes
  - Outline for the 2 days
  - Material and presentations
  - Online recording
  - Previous experience / current problems
    - LIST:
- **Introduction (8:00-9:00)**
  - Glacier melt and water availability
  - Groundwater vs ice and snow
  - Modelling equifinality
  - <u>Hydrological observations</u>
    - Water level
      - Semi-quantitative
    - Discharge
      - Quantitative
    - Water physical properties and chemistry
      - Elect. Conductivity
      - Temperature
      - Stable isotopes
      - Radioactive isotopes (129I)

- **Theoretical basis (9:15-10:00)**
  - **“How to get continuous discharge estimates?”**
  - Water level
    - Pressure transducers
    - Installation position
    - Parameters (installation, angle, depth, etc.)
  - Discharge (fluorometer)
    - Global method
  - Rating curve
    - Stream channel cross sections
    - Polynomial function
    - H0 – depth of no flow

- **Water level (10:00-11:00)**
  - Instruments
    - HOBO
    - Van Essen
    - Others ...
    - 2 x (atmosphere and water)
  - Construction/Installation
    - Position (water)
      - Pools
      - Rapids
      - Bolder sizes
      - Indicators for (un)stable riverbeds
    - Tubes and tools
      - Wire, drill, saw, bolts, threaded rods, ...
      - Tape measure
      - Angle measurement
      - GPS and photos
    - Verify depth of water probe
      - Photos, triangulation, tape measure
  - Starting devices & reading out data
    - Software
    - Shuttle/USB reading device

- **Discharge introduction (11:00-12:00)**
  - Fluorescence
  - Tracer
    - Toxic vs non-toxic
    - Detection limit
  - Optical method
    - Problems with sediments
  - Mixing length
  - Injection amount
  - Calibration
  - Color deterioration under sunlight

- **Discharge preparation (13:00-14:00)**
  - Instruments, equipment and tools
    - FL30
    - Logger

        - Battery
        - SD-card
    - Choosing location
    - Setup instrument
    - Calibration solution (preparation – before the fieldwork)

- **Discharge measurements (14:00-15:00)**
    - Calibration (in the field)
    - Injection (preparation – before the fieldwork)
    - Injection (in the field)
    - Repeated measurements
        - Water level variation
        - Switching between sites
        - Repeated calibration
        - Documentation (water quality, timing, weather, photos…)
    - Trouble shooting
        - Jumping signals
        - Unclear mixing /switching positions
    - Dangers (sun exposition)
        - Overestimation!

- **Programs introduction (15:00-15:30)**
    - Water level logger software
        - HOBOWare
        - Diver Office
        - Excel
    - Renkulab
        - Python
        - RStudio
        - Parameter file
    - Local installation (see installation tutorial)
    - Excel
- **Discussion (15:30-16:00)**

Day 2 ----------

- **Water level analysis (08:00-09:00)**
    - Reading data from logger
    - CSV file cleaning (Excel)

        - L2 – frozen and unreliable data removal
    - Atmospheric correction
    - Water base level adjustments (renkulab)
- **Discharge data preparation (09:00-10:00)**
    - Data from SD-card
    - Transformation binary to mV
    - Splitting individual measurements
    - Preparing metadata file (Excel)
        - Notebook records!

- **Discharge data analysis I (10:00-11:00)**
    - RScripts (interactive work)
    - Noise
    - Calibration
    - Breakthrough curve

- **Discharge data analysis I (11:00-12:00)**
    - Version 1 & Version 2
    - Injection amount
        - Base solution
        - Injection volume
        - (calibration amounts – Version 2)
    - Repeated measurements
    - With and without extrapolation (early measurement interruption)
    - Filling in the overview file (Excel)

- **Rating curve data preparation (13:00-14:00)**
    - Data pair (water level – discharge) preparation (Excel)
    - H0 range definition
    - Manual showcase polynomial fit
        - Uncertainty
        - H0
- **Rating curve calculation (14:00-15:00)**
    - Results
        - Uncertainty
        - Low, mid, high
    - Issues
    - Work in progress

- **Discussion (15:00-16:00)**
    - Feedback

- Future work in Central Asia
- Combined scientific research database
  - In prep.
  - Interest?